h2-watterleak-sensor rev A — ESP32-H2 Matter-over-Thread water-leak detector

Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Probe

File: probe.kicad_sch

UI_Prog

File: ui.kicad_sch

Blocks (one sheet each):
  Power   — CR123A, JP1 current jumper, Q1 reverse-polarity PFET, U2 TPS63900 3.3 V, battery divider
  MCU   — U1 ESP32-H2-WROOM-02C-H4S, EN RC, 32.768 kHz crystal, USB series resistors
  Probe   — channel A (2 bottom comb electrodes) + channel B (J2 remote probe), zero standby current
  UI_Prog — BOOT/user button, status LED, J1 programming header
Design notes: docs/03-so-do-nguyen-ly.md | Power budget: tools/power_budget.py

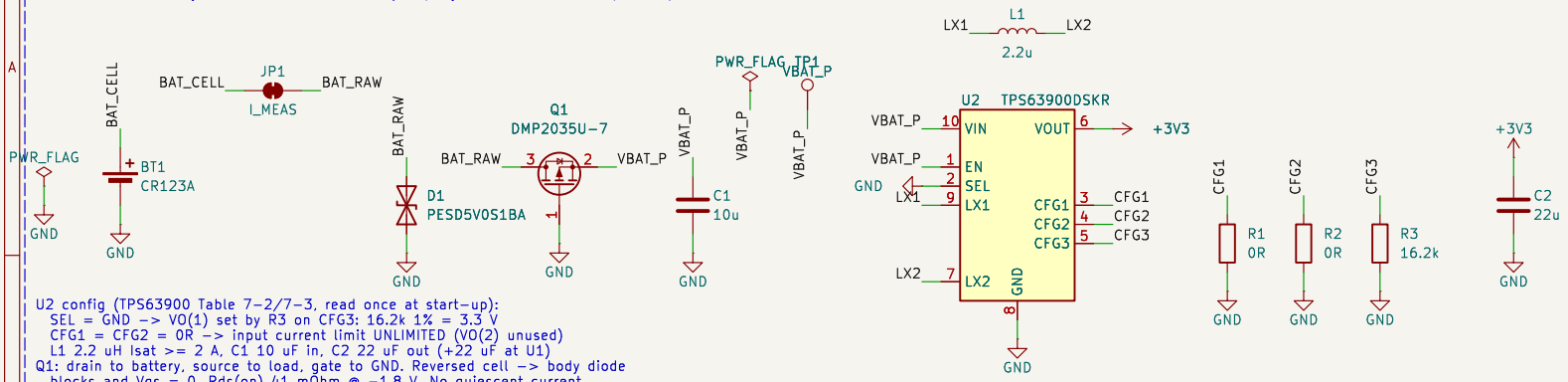
Generated by hardware/tools/gen_sch.py — edit the generator, not this file
ESP32-H2 Matter-over-Thread water-leak sensor (CR123A, ICD/SED)
h2-watterleak-sensor

Sheet: /
File: h2-leak-sensor.kicad_sch

Title: h2-watterleak-sensor — top level

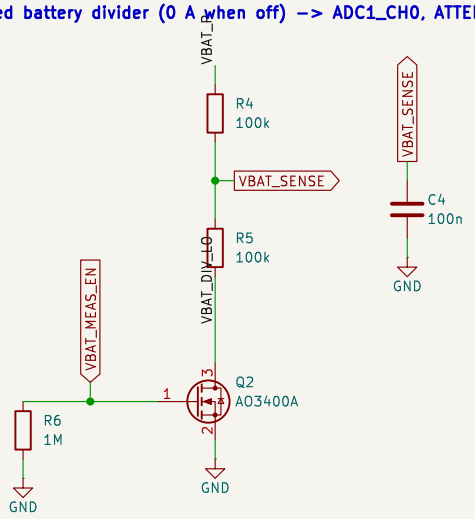
Size: A4	Date: 2026-11-12	**Rev: A**
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CR123A -> JP1 (current-measurement jumper) -> Q1 reverse-polarity PFET -> U2 TPS63900 buck-boost -> +3V3



U2 config (TPS63900 Table 7-2/7-3, read once at start-up):
SEL = GND -> VO(1) set by R3 on CFG3: 16.2k 1% = 3.3 V
CFG1 = CFG2 = 0R -> input current limit UNLIMITED (VO(2) unused)
L1 2.2 uH Isat >= 2 A, C1 10 uF in, C2 22 uF out (+22 uF at U1)
Q1: drain to battery, source to load, gate to GND. Reversed cell -> body diode blocks and Vgs = 0. Rds(on) 41 mOhm @ -1.8 V. No quiescent current.
JP1: bridged by default; cut and fit a 2-pin header to measure the sleep current.

Switched battery divider (0 A when off) -> ADC1_CH0, ATTEN2 (0..1.9 V)



VBAT_SENSE = VBAT_P / 2 when Q2 is on (VBAT_MEAS_EN = 1).
Q2 off -> no current path; VBAT_SENSE sits at VBAT_P (<= 3.3 V for CR123A) through R4.
Settling: tau = (R4||R5) * C4 = 5 ms -> wait >= 30 ms before sampling.
R6 keeps Q2 off while GPIO0 floats (reset, light-sleep with TOP power-down).
Divider current while measuring: 16.5 uA for ~30 ms, a few times per day.

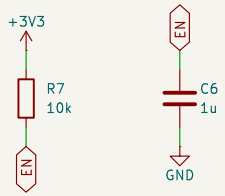
Generated by hardware/tools/gen_sch.py - edit the generator, not this file
ESP32-H2 Matter-over-Thread water-leak sensor (CR123A, ICD/SED)
h2-watterleak-sensor

Sheet: /Power/
File: power.kicad_sch

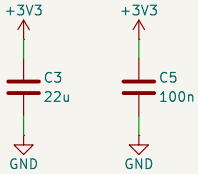
Title: Power: CR123A, protection, TPS63900 3.3 V, battery sense

Size: A4	Date: 2026-11-12	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/5

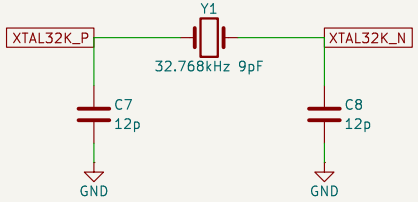
EN power-up delay (datasheet Fig. 5)



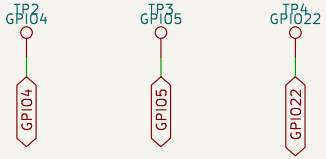
Decoupling at U1 pin 1

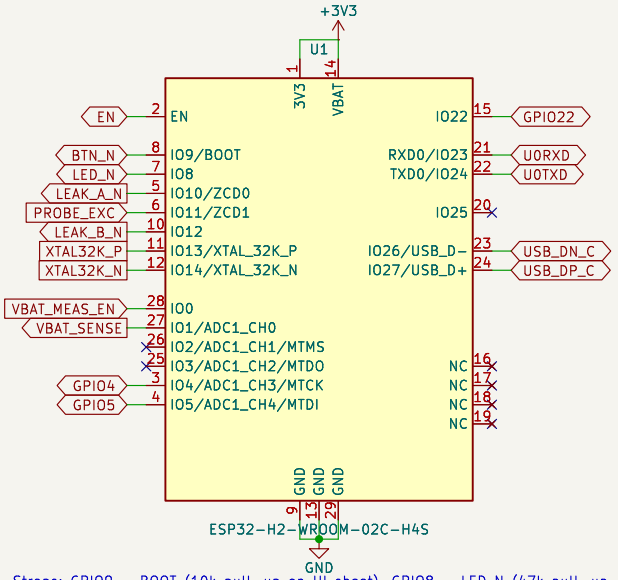


32.768 kHz RTC crystal (RTC_CLK_SRC_EXT_CRYS)



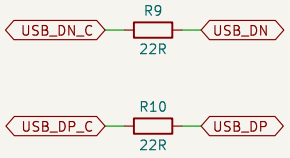
Spare GPIO test pads





Straps: GPIO9 = BOOT (10k pull-up on UI sheet), GPIO8 = LED_N (47k pull-up -> 1 at reset).
GPIO25 not connected (only used if eFuse STRAP_JTAG_SEL_ENABLE is burnt - do not burn it).
Wake sources: EXT1 ANY_LOW on GPIO9/10/12 (LP IO). PROBE_EXC = GPIO11 held in sleep.
Module VBAT pin is the 3.0-3.6 V RTC supply input -> tied to +3V3, never to the CR123A.

USB Serial/JTAG series resistors



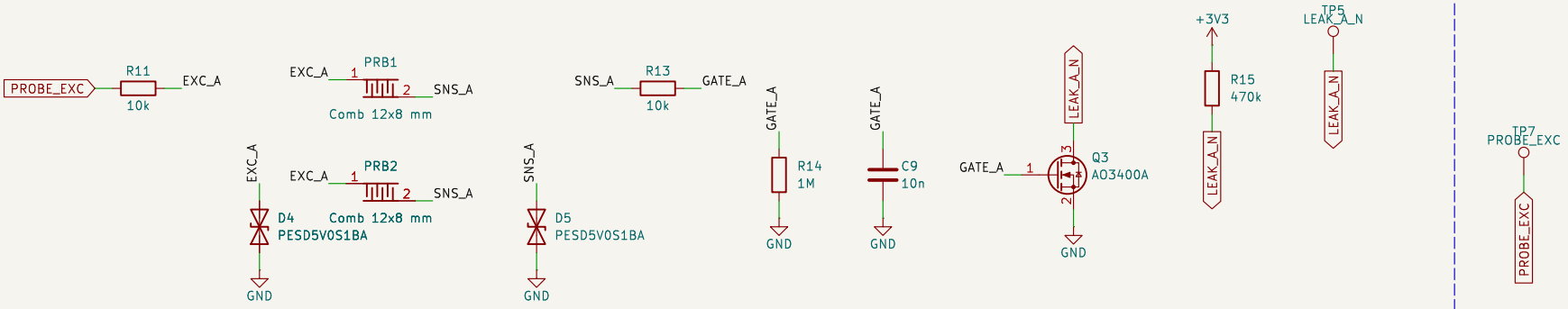
Generated by hardware/tools/gen_sch.py - edit the generator, not this file
ESP32-H2 Matter-over-Thread water-leak sensor (CR123A, ICD/SED)
h2-watterleak-sensor

Sheet: /MCU/
File: mcu.kicad_sch

Title: MCU: ESP32-H2-WROOM-02C-H4S, reset, 32.768 kHz, USB/UART

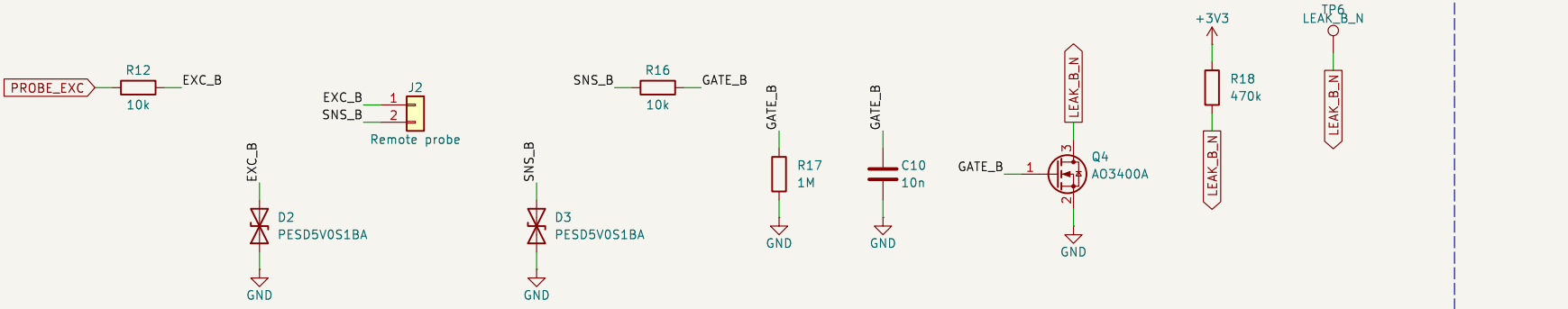
Size: A4	Date: 2026-11-12	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 3/5

Channel A — two comb electrodes on the PCB bottom (in parallel)



EXC_A: PROBE_EXC (GPIO11) through 10k. SNS_A -> 10k -> gate; 1M pull-down sets the trip point, 10 nF filters RF / splashes. Dry: open circuit, 0 A. Wet: Vg = 3.3 V * 1M / (1M + Rw + 20k) > Vth -> LEAK_A_N low -> EXT1 wake. Trip Rw ~ 1.3–4 MOhm (AO3400A Vth 0.65–1.45 V). Max DC through water 3.3 uA; firmware then drops PROBE_EXC and pulses it 5 ms every 10 s.

Channel B — remote probe on J2 (JST-PH 2 pin)



EXC_B: PROBE_EXC (GPIO11) through 10k. SNS_B -> 10k -> gate; 1M pull-down sets the trip point, 10 nF filters RF / splashes. Dry: open circuit, 0 A. Wet: Vg = 3.3 V * 1M / (1M + Rw + 20k) > Vth -> LEAK_B_N low -> EXT1 wake. Trip Rw ~ 1.3–4 MOhm (AO3400A Vth 0.65–1.45 V). Max DC through water 3.3 uA; firmware then drops PROBE_EXC and pulses it 5 ms every 10 s.

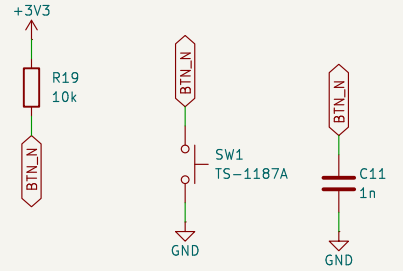
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ESP32-H2 Matter-over-Thread water-leak sensor (CR123A, ICD/SED)
h2-watterleak-sensor

Sheet: /Probe/
File: probe.kicad_sch

Title: Water probes: 2 channels, zero standby current

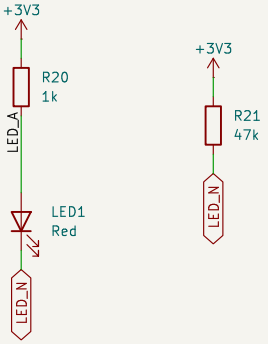
Size: A4	Date: 2026-11-12	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 4/5

Button: BOOT strap + user (EXT1 wake)



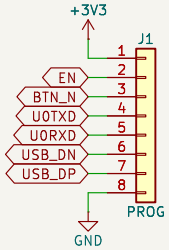
Short press: identify / status. Hold 10 s: factory reset.
Hold while EN rises: download mode (GPIO9 = 0, GPIO8 = 1).
Keep the RC small: tau = 10k||45k * 1 nF = 8 us << t_H 3 ms.

Status LED (active low on GPIO8 strap)



I_LED = (3.3 − 1.9) / 1k = 1.4 mA (blinks only).
R21 keeps GPIO8 = 1 at reset (download-boot strap)
and keeps the LED off while GPIO8 is hi-Z in sleep.

J1 programming / pogo-jig header (fit header or use pogo pins)



J1 pin-out: 1 +3V3 (jig supplies 3.3 V, battery removed), 2 EN, 3 IO9/BOOT,
4 TXD0 (GPIO24), 5 RXD0 (GPIO23), 6 USB D− (GPIO26), 7 USB D+ (GPIO27), 8 GND.
USB Serial/JTAG: esptool can enter download mode by itself.
UART: the jig drives EN and IO9 (DTR/RTS auto-reset circuit lives on the jig).

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ESP32-H2 Matter-over-Thread water-leak sensor (CR123A, ICD/SED)
h2-watterleak-sensor

Sheet: /UI_Prog/
File: ui.kicad_sch

Title: User interface and programming header

Size: A4	Date: 2026-11-12	**Rev: A**
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